

Project ID:

25-26J-423

1. Topic (12 words max)

Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka

2. Research group the project belongs to

MR - Mixed Reality

3. Specialization of the project belongs to

Interactive Media (IM)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The Sri Lankan government has decided to make **History** an elective subject in middle school, though the change is not yet implemented. Many believe History is essential for preserving cultural heritage, national identity, and critical thinking skills. Prioritizing STEM subjects risks further disconnecting students from their historical roots.

Most schools still depend on traditional, textbook-based teaching, encouraging memorization over understanding. Science and Mathematics remain abstract without practical work, while English is often taught without real-life application. Rural schools face even greater challenges due to limited resources.

Virtual Reality (VR), especially when combined with **gamification**—points, missions, and rewards—can turn passive learning into immersive, engaging experiences. Yet in Sri Lanka, VR tools are rarely syllabus-aligned, interactive, or available in local languages.

Our project proposes a **gamified VR platform** integrating History with Mathematics, Science, and English for Grades 7–9. The game progresses through major historical kingdoms of Sri Lanka, with each level offering a narrated introduction, immersive 3D exploration, and syllabus-based challenges tailored to the student's grade and age.

This approach ensures that even if History becomes elective, students will still engage with the nation's past while strengthening essential academic skills.

- [1] J. Radianti, T. A. Majchrzak, J. Fromm, and I. Wohlgenannt, "A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda," *Computers & Education*, vol. 147, p. 103778, 2020. [Online]. Available: <https://doi.org/10.1016/j.compedu.2019.103778>
- [2] J. Parong and R. E. Mayer, "Learning science in immersive virtual reality: A case study of student engagement," *Journal of Educational Psychology*, vol. 110, no. 6, pp. 785–797, 2018. [Online]. Available: <https://doi.org/10.1037/edu0000241>
- [3] C. Gunawardena and S. Karunanayake, "E-learning and Distance Education in Sri Lanka: A Policy and Practice Perspective," *The International Review of Research in Open and Distributed Learning*, vol. 15, no. 2, pp. 185–210, 2014. [Online]. Available: <https://doi.org/10.19173/irrodl.v15i2.1679>
- [4] A. Kushniruk and E. Borycki, "Integrating Virtual Reality (VR) Into Education: Theory and Practice," in *International Conference on e-Learning*, 2021, pp. 123–132. [Online]. Available: <https://www.ejel.org>
- [5] M. M. A. C. D. Perera, N. C. M. Bandara, and L. T. P. D. Piyathilaka, "Study on Impact of Virtual Reality in Learning Environment with Theoretical Perspectives," in *Proc. Int. Conf. Adv. ICT Emerging Reg.*, 2019, pp. 1–6. [Online]. Available: <https://doi.org/10.1109/ICTER48817.2019.9023754>
- [6] P. Jayatilleke and G. Gunawardena, "Enhancing Education through Gamification: Case Study in Sri Lankan Context," in *Proc. Int. Conf. Interact. Collab. Learn. (ICL)*, 2020, pp. 35–40. [Online]. Available: <https://doi.org/10.1109/ICL.2020.9248701>

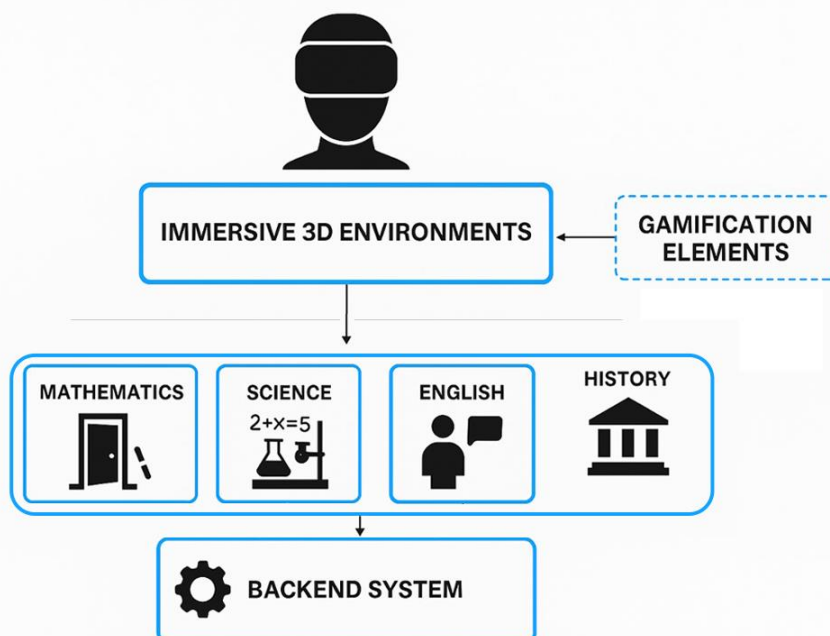
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This project proposes a **gamified Virtual Reality (VR) educational application** designed for Sri Lankan middle school students (Grades 7–9). The platform integrates **Mathematics, Science, English, and History** into a single immersive experience aligned with the **national curriculum**. The solution runs on **Meta Quest (Oculus) headsets** using the Unity engine and leverages game mechanics to improve student engagement and learning outcomes.

Each level of the VR experience is based on a significant **Sri Lankan historical scene** (e.g., Anuradhapura period, colonial Ceylon). Students first encounter a **short narrative introduction** to the historical context, delivered through guided storytelling and 3D environments. They then complete **subject-specific tasks**—math puzzles, science simulations, and English conversations—embedded within that historical environment. These tasks are created according to the official syllabus to reinforce classroom learning through applied interaction.

Gamification features such as **points, levels, timers, and digital badges** motivate learners, while a **backend system** tracks their progress and provides feedback to teachers. The app supports **Sinhala and English**, ensuring inclusivity and wider accessibility, particularly for students in under-resourced regions.

CONCEPTUAL DIAGRAM



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project requires expertise in **VR development, interactive media, game design, and curriculum-based education**. The team uses **Unity and C#** to build immersive 3D environments and interactive subject modules optimized for **Meta Quest** headsets.

Core skills include **3D modeling, animation, VR optimization, and gamification** techniques such as points, levels, and adaptive difficulty. These are applied to create engaging educational tasks aligned with Sri Lanka's Grade 7–9 syllabus in **Math, Science, English, and History**.

Educational design knowledge is essential to ensure age-appropriate, syllabus-mapped learning experiences. Language expertise supports the development of **Sinhala–English dual-language content**, including subtitles and voiceovers.

Data Requirements:

- National curriculum and textbooks (Grades 7–9)
- Past papers and activity sheets
- Teacher and student feedback (via surveys/interviews)
- Open-source 3D assets and historical references
- User test data for evaluating performance and usability

This combination of technical and educational knowledge ensures a well-rounded, interactive VR learning tool tailored to Sri Lankan classrooms.

8. Objectives and Novelty

Main Objective To develop a gamified Virtual Reality (VR) educational application for Sri Lankan students in grades 7-9, delivering interactive and syllabus aligned content in Mathematics, Science, English, and History. The application will use immersive 3D environments and gamification elements to enhance student engagement, learning effectiveness, and long-term retention through subject specific, mission-based VR learning.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Dayananda W.C.R IT22561602	Design and implement VR-based Mathematics modules aligned with curriculum outcomes	- Analyze Grade 7-9 Mathematics syllabus. - Develop VR-based interactive lessons (algebra, geometry). - Create problem-solving environments using triggers (open a gate by solving an equation). - Integrate adaptive difficulty. - Include scoring and reward systems for each completed math challenge.	Applies gamified mathematical logic to improve conceptual clarity using interactive VR experiences tailored to local curriculum
Wijenarayana C.H IT22132314	Develop immersive Science learning environments with virtual labs and natural simulations	- Extract key Science topics (plants, ecosystems, motion) from the curriculum. - Build safe virtual laboratories simulating experiments (photosynthesis, electricity flow) - Include animated, step-by-step guidance	Offers practical science experience to students in under resourced schools by replicating experiments in an immersive and risk-free virtual environment

		• Allow students to interact with lab tools in VR • Enable feedback after completion of each experiment	
Jayalath K.M.K.D IT22581570	Build gamified History timelines and cultural scenarios with interactive storytelling	• Map historical events in Grades 7-9 syllabus (Anuradhapura period, colonialism) • Recreate 3D historical environments and key landmarks • Embed guided narration, timeline interactions, and mission-based learning (complete a task as an ancient king or citizen) • Implement cultural storytelling and historical quizzes	Brings history to life through cultural immersion and narrative-driven missions, allowing students to "experience" events instead of memorizing them
Ubeywana R.C IT22339874	Create dual-language (Sinhala–English) English learning missions focused on conversation, grammar, and comprehension.	• Identify language learning objectives in the curriculum • Create VR-based English learning quests (role-playing as a shopkeeper or tourist) • Design vocabulary games and sentence construction challenges • Integrate subtitles, voiceovers, and interactive feedback • Include real-life settings like markets, classrooms, or stations to enhance practical use of English	Promotes language acquisition through interactive storytelling and bilingual scenario-based tasks; supports speech and listening practice via real-time VR

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Dayananda W.C.R IT22561602	Specializes in immersive learning experience design by creating VR-based mathematics environments that visualize abstract concepts such as algebra and geometry. Using Unity and C#, the module implements interactive object manipulation (e.g., triggering gates by solving equations), mission-based progress, adaptive difficulty, and feedback loops. This aligns with the Interactive Media specialization by applying spatial interaction , real-time responsiveness , and gamified logic modeling to enhance student comprehension in math.
Wijenarayana C.H IT22132314	Focuses on scientific simulation and visualization through the development of virtual laboratories and natural ecosystems in VR. Unity's physics engine and animation systems are used to replicate hands-on experiments (e.g., photosynthesis, electricity flow) in a risk-free environment. This work complies with the Interactive Media specialization by integrating simulation-based learning , interactive storytelling , and physics-driven interactivity to create realistic, engaging science experiences that promote inquiry-based learning.
Jayalath K.M.K.D IT22581570	Emphasizes experiential storytelling and cultural immersion by recreating historical timelines and events in 3D VR environments. Using Unity's animation system and timeline controls, students interact with characters and landmarks in mission-based story arcs. The component applies Interactive Media principles such as narrative design , experiential learning , and behavioral engagement to allow students to "live through" history, rather than memorize it, thereby fostering deeper historical understanding.
Ubeywana R.C IT22339874	Centers on dual-language (Sinhala–English) interactive communication, enabling students to practice vocabulary, grammar, and speaking through VR-based roleplay (e.g., acting as a shopkeeper or tourist). The integration of subtitles, voiceovers, and real-life conversation scenarios reflects core Interactive Media techniques such as multimodal interaction , context-based learning , and real-time feedback systems . This supports both language acquisition and confidence-building through immersive storytelling and dialogue.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Didula Chamara	Thanaweera Arachchi	
Co-Supervisor	Mr.	Nushkan	Nismi	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

[illegible]

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.